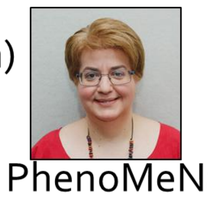
Metabolomics Community-led Implementation Study: Standardizing the Fluxomics Workflows

ELIXIR All Hands , 7-10 June 2022, Amsterdam, Netherlands

SHORTH DESCRIPTION

Fluxomics is an integral part of the bioinformatics and systems biology toolbox, however its broader use and integration with other omics is currently challenged by lack of: (i) standardized isotopic labeling data repositories, (ii) tool interoperability and (iii) harmonized fluxomics training workflows. In this context, ELIXIR Metabolomics Community decided to focus its 2nd implementation study (IS) on the standardization of fluxomics workflows. This IS ran from 06/2019 to 11/2021 and involved mainly three ELIXIR Platforms (Tools, Interoperability, Training), being in alignment with Data & Compute. It was supported by EMBL-EBI and thirteen national ELIXIR nodes, co-coordinated by ELIXIR-GR and ELIXIR-ES with management support by ELIXIR-NL.

Management Team & Partners

- **Leading Partners**: ELIXIR-GR (FORTH /M. Klapa) 

ELIXIR-ES (UB/M. Cascante) – PhenoMeNal fluxomics

- **Management**: ELIXIR-NL (M. van Rijswijk)
- 14 ELIXIR Nodes Participating
- 3 Platforms (Interoperability, Training, Data)
- **Tools**/Compute related activities included too
- Aligned with Communities : Human Data, Plant Sciences, **Proteomics**, Galaxy, **Microbial Biotech**, Toxicology, Food & Nutrition, **Systems Biology** (emerging)

Objectives & Outcomes

Tools-Related Activities

Objective T1: Summarize the (open source) tools used in fluxomics research and their characteristics, encourage their inclusion to the bio.tools registry, as well as make a first inventory of the (interoperability) challenges of these tools.

Deliverable T1: Carried out a survey about tools and interoperability needs ✔
in collaboration with ELIXIR Tools – included fluxomics tool directory in bio.tools registry

Objective T2: Add the most important tools in fluxomics to the ELIXIR bio.tools registry, by organizing a hackathon **in collaboration with the Tools platform**.

Deliverable T2: Hackathon in Toulouse, January 20-21 2020

✔ Discussed the results of the survey – interoperability issues
Collected all fluxomics-related tools in bio.tools registry under the fluxomics directory

Data-Related Activities

Objective D1: Perform a gap analysis on standards for isotopic labeling data deposition and based on this, develop a standardized repository, accordingly extending the MetaboLights database (EMBL-EBI). ✔

Deliverable D1: Finalized a proposal for standards for the FAIRified deposition of isotopic labeling data and the way MetaboLights Database (EMBL-EBI) should be accordingly extended (lead ELIXIR-UK (P. Rocca-Serra, UOXF), with EMBL-EBI MetaboLights Database, contribution by ELIXIR-FR and ELIXIR-ES) ✔

Deliverable D2: Initiated implementation of this standardized deposition system to the MetaboLights Database (lead: EMBL-EBI MetaboLights Database with ELIXIR-UK). ✔

Interoperability-Related Activities

Deliverable I1: Extended PhenoMeNal flux software to LC-MS and NMR isotopic data & appropriately adapt Isodyn software (lead: ELIXIR-ES).

Deliverable I2:
(a) Consolidated workflow outputs towards standard compliant files (SBML, JSON data packages), lead ELIXIR-UK (UOXF);
(b) Extended OpenMS to 13C-fluxomics data, lead ELIXIR-DE (U Tübingen);
(c) Added IsoCor and Influx_si to the workflow, lead ELIXIR-FR (INRA-MetaboHub);
(d) Comparative testing of tools for large-scale 13C –based flux analysis; adding tool for atomically resolve a metabolic reconstruction, lead ELIXIR-NL (UL)

Deliverable I3:
(a) Extended *BridgeDb resource* for reaction identifier mapping into fluxomics (lead: Interoperability platform located at ELIXIR-NL, C. Evelo//E. Willighagen, Umaas)

Training-Related Activities

Deliverable Tr1: Made an inventory of fluxomics-related training activities and materials (leads: ELIXIR-GR & ELIXIR-ES)

Deliverable Tr2: Organised a 2-part hackathon (Feb & Mar 2021) to prepare the programme and course materials for the 1st ELIXIR Fluxomics Training School in Oct 2021 (leads: ELIXIR-GR and ELIXIR-ES)

Deliverable Tr3: Organised the 1st (virtual, lead: ELIXIR-GR)

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